

Introduction

Large Language Models (LLMs) are increasingly used to answer requests based on provided documents. Yet, they often rely on memorized facts, or try to retrieve parts of the text without genuine reasoning as to whether it answers the request (Du et al., 2025). A way to improve LLMs capability to answer their prompt based on provided documents is to perform a post-training reinforcement learning (RL) step using Question-Answer (QA) datasets.

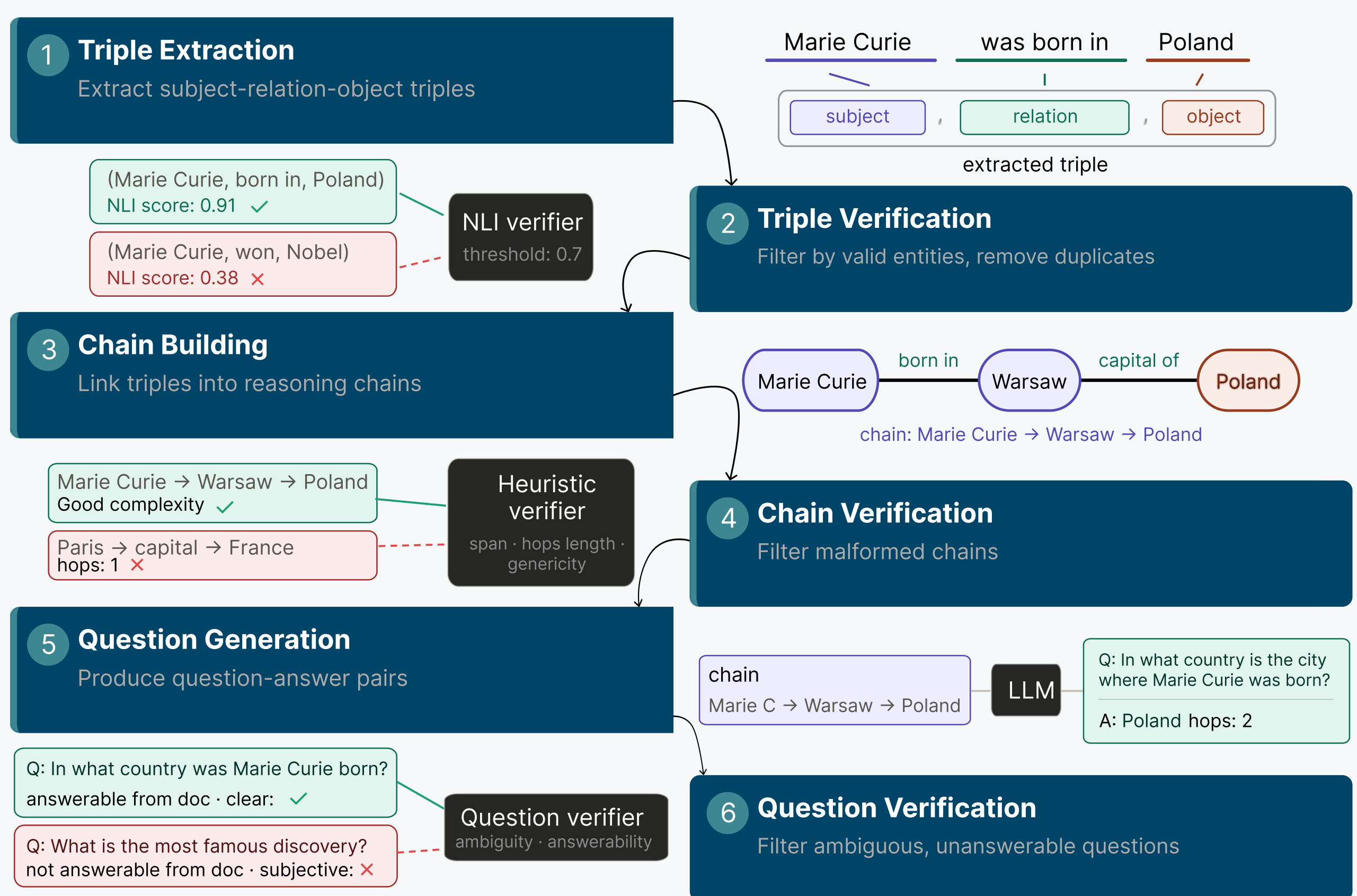
However, most existing QA datasets tend to contain only small, easily answerable hand-picked QA pairs.

Contributions

We make the following contributions:

- **Automated multi-hop QA dataset pipeline:** an end-to-end system combining NLP and ML techniques to extract large-scale, long-context question-answer pairs requiring multi-step reasoning.
- **Filtering methods:** A set of filtering methods to ensure factual grounding, answer verifiability, and reasoning chain coherence.
- **LLM post-training validation:** empirical verification of the dataset's effectiveness by post-training an LLM using multiple RL methods.

Automated Dataset Creation Pipeline



Example: Q: "Which valley is the destination of the southeastward drainage from the elevated western region of Rhode Island?"
→ A: "Connecticut Valley"

Policies

All policies we tested are variants of classic algorithms (PPO, REINFORCE) that estimate advantages without a learned value function, reducing computational need due to the critic model.

■ GRPO (Shao et al., 2024)

Based on PPO, bypasses the critic by sampling G responses per prompt and normalizing rewards within the group:

$$\hat{A}_i = \frac{r(x, y_i) - \text{mean}(\{r(x, y_i)\})}{\text{std}(\{r(x, y_i)\})}$$

■ GSPO (Zheng et al., 2025)

Fixes GRPO's variance and instability by replacing token-level importance ratio by a **sequence-level** one, via the importance ratio

$$s_i(\theta) = \exp\left(\frac{1}{|y_i|} \sum_{t=1}^{|y_i|} \log \frac{\pi_{\theta}(y_{i,t}|x, y_{i,<t})}{\pi_{\theta_{\text{old}}}(y_{i,t}|x, y_{i,<t})}\right)$$

■ RLOO (Ahmadian et al., 2024)

Based on REINFORCE, computes the baseline for rollout i using the $G - 1$ other samples:

$$\hat{A}_i = r_i - \frac{1}{G-1} \sum_{j \neq i} r_j$$

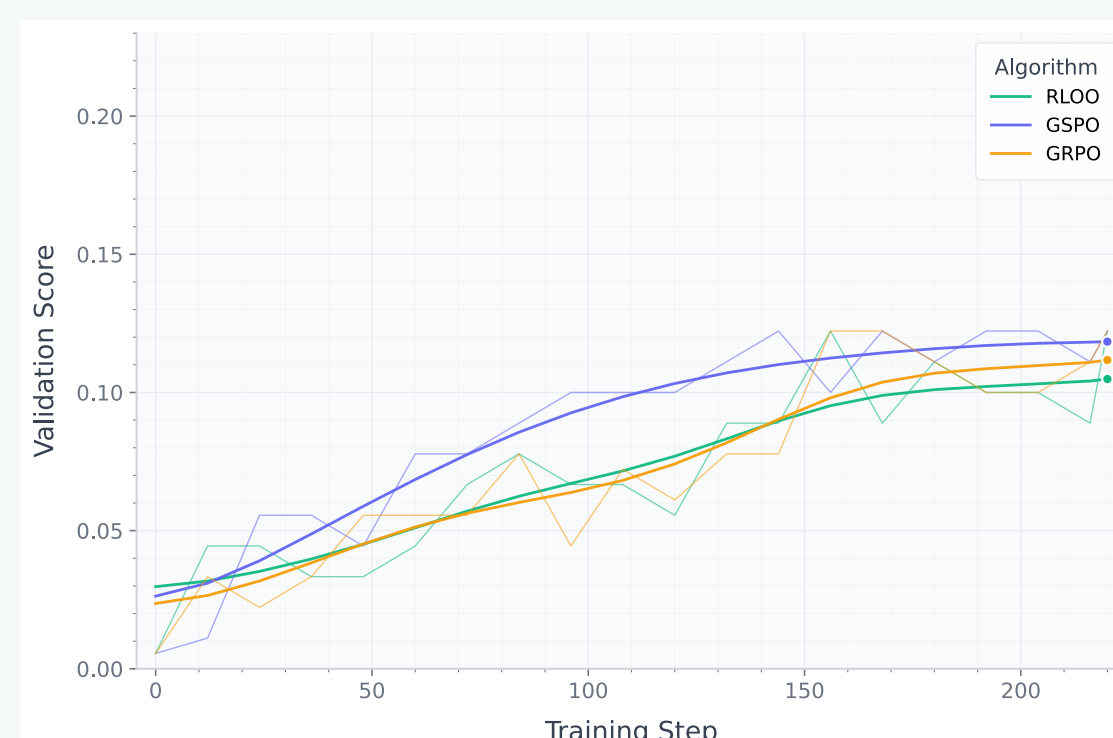
Future Work

To improve on this project:

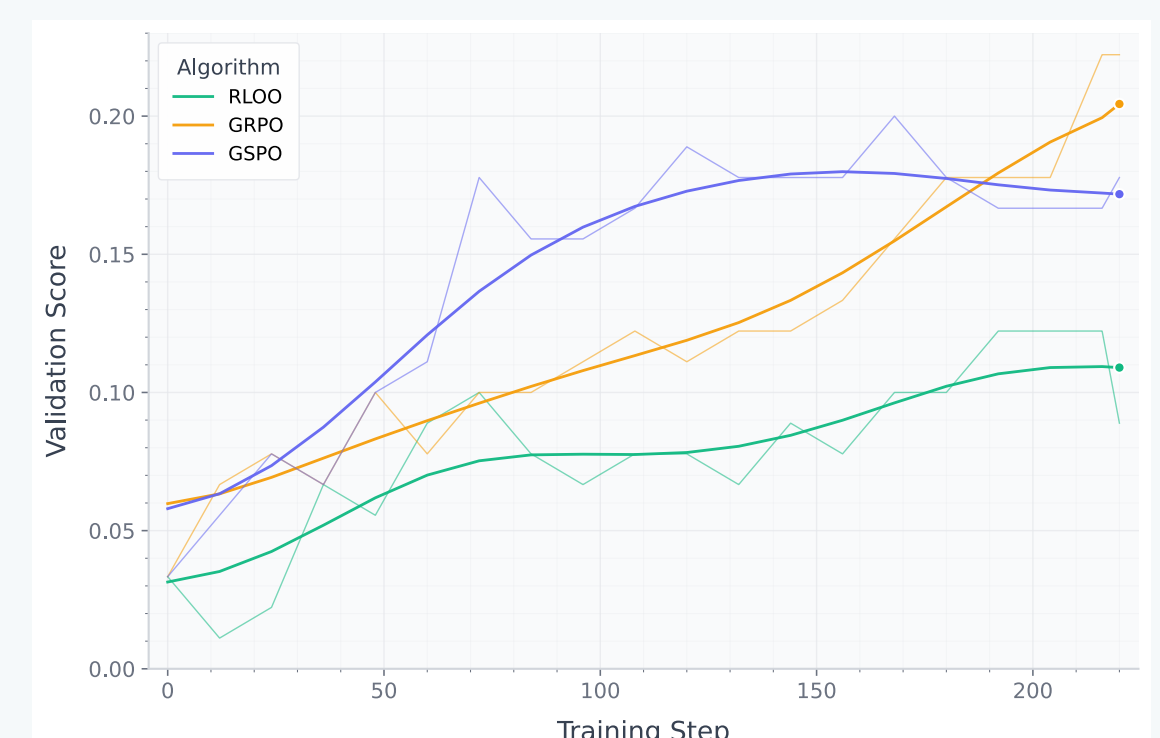
- **Ablation study:** systematically evaluate each pipeline stage to quantify their individual contribution.
- **Scale validation:** post-train on a larger dataset to confirm reasoning gains.
- **Filtering:** enforce stricter constraints, and add semantic de-duplication to the pipeline.
- **Hyperparameter search:** conduct a systematic search over RL parameters to identify optimal training configurations.
- **Apply to Apertus:** the generated datasets will finally be used to post-train Apertus.

RL post-training

We chose Qwen2.5-7B-Instruct (Qwen et al., 2025) and trained it for 20 epochs using vERL. Models are initially unable to answer questions, but can improve, validating our process. Results below show that training is strongly influenced by reward function design and prompt:



(a) Validation score - Baseline reward



(b) Validation score - Custom reward

Conclusion

We successfully built an end-to-end automated pipeline for multi-hop QA dataset generation, producing over 450 QA pairs using **NLP** extractor and over 650 QA using the **Hybrid** extractor, without the need for human annotation. Despite the limited scale, post-training experiments across three RL algorithms demonstrate consistent improvements in long-context reasoning. These results validate the pipeline design, confirm that our dataset is challenging for state-of-the-art models and suggest that scaling to larger corpora and improving the filtering stages are the most impactful directions for future work.

References

- Ahmadian, A., Cremer, C., Gallé, M., Fadaee, M., Kreutzer, J., Pietquin, O., Üstün, A., & Hooker, S. (2024). Back to basics: Revisiting reinforce style optimization for learning from human feedback in llms. <https://arxiv.org/abs/2402.14740>
- Du, Y., Tian, M., Ronanki, S., Rongali, S., Bodapati, S., Galstyan, A., Wells, A., Schwartz, R., Huerta, E. A., & Peng, H. (2025). Context length alone hurts llm performance despite perfect retrieval. <https://arxiv.org/abs/2510.05381>
- Qwen, : Yang, A., Yang, B., ..., & Qiu, Z. (2025). Qwen2.5 technical report. <https://arxiv.org/abs/2412.15115>
- Shao, Z., Wang, P., Zhu, Q., Xu, R., Song, J., Bi, X., Zhang, H., Zhang, M., Li, Y. K., Wu, Y., & Guo, D. (2024). Deepseekmath: Pushing the limits of mathematical reasoning in open language models. <https://arxiv.org/abs/2402.03300>
- Zheng, C., Liu, S., Li, M., Chen, X.-H., Yu, B., Gao, C., Dang, K., Liu, Y., Men, R., Yang, A., Zhou, J., & Lin, J. (2025). Group sequence policy optimization. <https://arxiv.org/abs/2507.18071>